

Mathematics for Computing

Assignment

Aleena Mary John Department of Computer Science and Engineering
with Artificial Intelligence and Software Engineering

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Set-Builder Form — Examples (1–5)

- ① Even numbers:

$$\{x \mid x = 2n, n \in \mathbb{Z}\}$$

- ② Natural numbers less than 10:

$$\{x \mid x \in \mathbb{N}, x < 10\}$$

- ③ Vowels in English:

$$\{x \mid x \text{ is a vowel in the English alphabet}\}$$

- ④ Perfect squares up to 100:

$$\{x \mid x = n^2, n \in \mathbb{Z}, 1 \leq n \leq 10\}$$

- ⑤ Odd integers:

$$\{x \mid x \in \mathbb{Z}, x = 2k + 1, k \in \mathbb{Z}\}$$

Set-Builder Form — Examples (6–10)

- 6 Primes less than 20:

$$\{x \mid x \text{ is prime, } x < 20\}$$

- 7 Multiples of 5 between 1 and 50:

$$\{x \mid x = 5n, n \in \mathbb{Z}, 1 \leq 5n \leq 50\}$$

- 8 Reals between 0 and 1:

$$\{x \mid x \in \mathbb{R}, 0 < x < 1\}$$

- 9 Days of the week:

$$\{x \mid x \text{ is a day of the week}\}$$

- 10 Solutions of $x^2 - 4 = 0$:

$$\{x \mid x \in \mathbb{R}, x^2 - 4 = 0\} = \{-2, 2\}$$

Types of Sets — AI/DS Examples (I)

- **Empty (Null) Set** — has no elements ($\emptyset$).
Predictions with model confidence $> 100\%$.
- **Singleton Set** — exactly one element.
Unique optimal weight vector for a strictly convex loss.
- **Finite Set** — countable, ends.
Class labels: $\{\text{spam}, \text{not spam}\}$.
- **Infinite Set** — doesn't end.
All possible real-valued weights of a neuron.
- **Equal Sets** — same elements, any order.
Two feature-selection runs agreeing on selected features.

Types of Sets — AI/DS Examples (II)

- **Equivalent Sets** — same cardinality.
Train and validation sets, both of size 1,000.
- **Subset** — every element of A is in B .
Positive predictions $\subseteq$ all predictions.
- **Universal Set** — contains everything relevant.
Full vocabulary in an NLP tokenizer.
- **Power Set** — set of all subsets.
All candidate feature subsets in wrapper-method selection.
- **Disjoint Sets** — no common elements.
Training set and test set (properly split, no overlap).

Inclusion–Exclusion Principle

For two sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Add both sets, then subtract the overlap once, since it was counted twice.

For three sets:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

General pattern: add all single sets, subtract all pairwise intersections, add back all triple intersections, and so on — alternating signs as you go deeper.

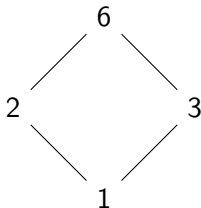
DS use case: counting users who clicked at least one of three ad campaigns, given the overlaps between campaigns.

Hasse Diagram of a Poset

Poset: a set with a relation that is reflexive, antisymmetric, and transitive — not every pair of elements needs to be comparable.

Hasse Diagram: draws only the covering relations (direct, non-redundant “less than” links), with greater elements placed higher. No arrows or self-loops are needed.

Example: $(\{1, 2, 3, 6\}, |)$ — “divides”



No line is drawn directly from 1 to 6, since that relation is already implied by $1 \rightarrow 2 \rightarrow 6$ and $1 \rightarrow 3 \rightarrow 6$.